



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VMS FUELS UNIT II

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

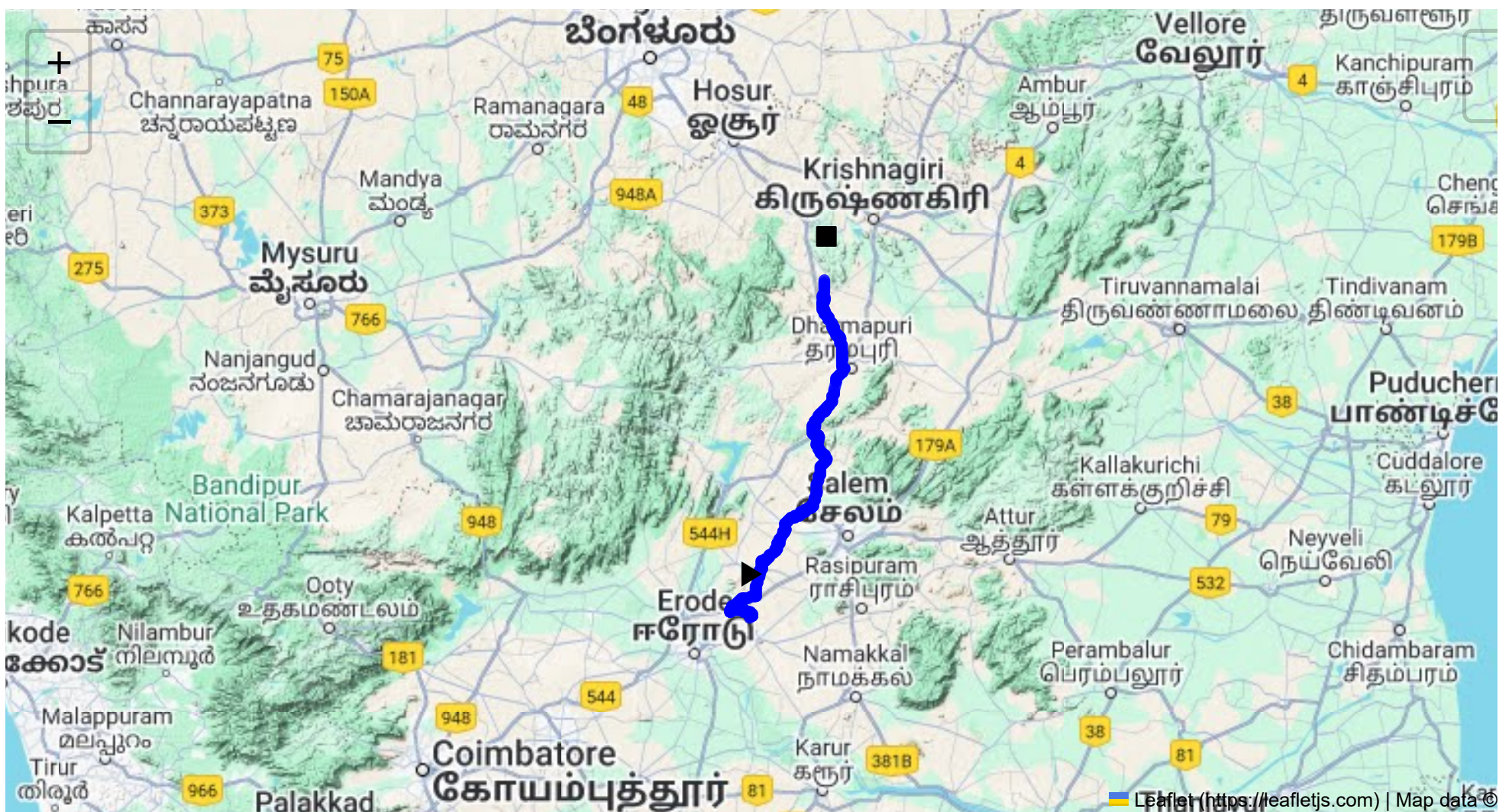
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 131.52 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.4 hours
Start: (11.4381, 77.8734)
End: (12.359966, 78.081809)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to 935J+VWV, Vellichandai spans approximately 131.52 kilometers, passing through multiple locations on significant roads like SH 86. It links several towns, providing a varied driving experience across urban, semi-urban, and rural areas.

Weather Conditions and Hazards

Tamil Nadu's climate is predominantly tropical. Common weather-related hazards include:

- **Monsoon Season (June to September):** Heavy rainfall can lead to waterlogging and reduced visibility.
- **Summer Heat (March to May):** Extreme temperatures might affect vehicle performance.

Advising extra caution during the monsoon for potential slippery roads and reduced visibility is crucial.

Traffic Patterns

- **Congestion Areas:** The towns of Sankari and Tharamangalam are likely points for congestion, especially during local markets or festivals.
- **Peak Hours:** Typically between 8:00 AM - 10:00 AM and 4:30 PM - 7:00 PM. These can extend in urban areas due to school and work traffic.

Road Quality and Infrastructure

- **SH 86:** Generally well-maintained but may have occasional potholes or narrow stretches, especially in rural transitions.
- **Secondary Roads:** Risk of lesser maintenance, irregular pavements, and obstructions like parked vehicles.

Ensure regular checks on heavy vehicles to prevent breakdowns in poorly maintained areas.

Alternative Routes

For emergencies, consider rerouting through:

- NH44 for more consistency in road quality and available facilities.
- Local alternative roads after consultation with a navigation system or local guidance, bearing in mind the possibility of lesser road quality.

Local Regulations for Hazardous Material Transport

- Adherence to designated routes for hazardous materials, often SH and NH roads.
- Mandatory safety equipment and appropriate signage on vehicles.
- Authorization papers and emergency response procedures should be in place.

Historical Incidents

- Sparse data on historical incidents involving hazardous material in this region, indicating satisfactory safety records.
- Remaining updated on local news and official transport advisories is recommended.

Environmental Considerations

- **Protected Areas:** No significant environmental protection zones noted along the primary route.
- **Rural Sensitivities:** Watch for livestock crossings in rural sections and respect local agricultural areas.

Communication Coverage

- Generally good coverage in towns, with possible dead zones in rural stretches.
- Advisable to have backup communication means (e.g., satellite phone) especially for emergency use.

Emergency Response Times

- Urban areas: Estimated 15-30 minutes due to proximity to services.
- Rural areas: 30-60 minutes or more, depending on remoteness and local emergency infrastructure.

Overall Risk Assessment

This route, while traversing varied terrains, follows primarily well-trafficked highways, minimizing risk when proper precautions are taken. However, vulnerable sections during adverse weather and peaks in local traffic demand attentive driving and readiness for rerouting. Operational protocols must account for potential delays, communication challenges, and regulatory compliance to maintain safety standards effectively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45571, 77.81702	30 KM/Hr
8	Turn	High	11.45615, 77.81677	15 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57649, 77.90610	30 KM/Hr
13	Turn	Medium	11.59214, 77.90656	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	Medium	11.63833, 77.94818	30 KM/Hr
15	Turn	Medium	11.64171, 77.95184	30 KM/Hr
16	Turn	Medium	11.64243, 77.95198	30 KM/Hr
17	Turn	Medium	11.64721, 77.95118	30 KM/Hr
18	Turn	Medium	11.69559, 77.97112	30 KM/Hr
19	Turn	Medium	11.69806, 77.97537	30 KM/Hr
20	Turn	Medium	11.71386, 77.99552	30 KM/Hr
21	Turn	Medium	11.71735, 78.00019	30 KM/Hr
22	Turn	Medium	11.72765, 78.02291	30 KM/Hr
23	Turn	Medium	11.73786, 78.03505	30 KM/Hr
24	Turn	High	11.73652, 78.03866	15 KM/Hr
25	Turn	Medium	11.74257, 78.04708	30 KM/Hr
26	Turn	Medium	11.86805, 78.08353	30 KM/Hr
27	Turn	Medium	11.87022, 78.08380	30 KM/Hr
28	Turn	Medium	11.87913, 78.08206	30 KM/Hr
29	Turn	Medium	11.92566, 78.06503	30 KM/Hr
30	Turn	Medium	11.93699, 78.05412	30 KM/Hr
31	Turn	Medium	11.95551, 78.04706	30 KM/Hr
32	Turn	Medium	11.95810, 78.04971	30 KM/Hr
33	Turn	Medium	11.95865, 78.04991	30 KM/Hr
34	Turn	Medium	11.95963, 78.04954	30 KM/Hr
35	Turn	Medium	11.96072, 78.04963	30 KM/Hr
36	Turn	Medium	11.96281, 78.05124	30 KM/Hr
37	Turn	Medium	11.96384, 78.05141	30 KM/Hr
38	Turn	Medium	11.96442, 78.05104	30 KM/Hr
39	Turn	High	11.96557, 78.04930	15 KM/Hr
40	Turn	High	11.96657, 78.04930	15 KM/Hr
41	Turn	Medium	12.12041, 78.13284	30 KM/Hr
42	Turn	High	12.26533, 78.08428	15 KM/Hr
43	Turn	High	12.26550, 78.08440	15 KM/Hr
44	Turn	Medium	12.27173, 78.08098	30 KM/Hr
45	Turn	High	12.29579, 78.07370	15 KM/Hr
46	Turn	Medium	12.29681, 78.07576	30 KM/Hr
47	Turn	Medium	12.31143, 78.07861	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
48	Turn	Medium	12.31224, 78.07819	30 KM/Hr
49	Turn	High	12.31368, 78.07448	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
3	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
7	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
8	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
9	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
10	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
11	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
12	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
14	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
15	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
18	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
19	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
20	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
21	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
22	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
23	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
24	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
25	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
26	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
28	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
30	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
31	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
32	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
34	hospital	Government Hospital Palacode	12.2964824, 78.0761735	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
13	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
27	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
29	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
33	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57649, 77.90610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59214, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63833, 77.94818



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64171, 77.95184



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64243, 77.95198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69559, 77.97112



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69806, 77.97537



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71386, 77.99552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71735, 78.00019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72765, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73786, 78.03505



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73652, 78.03866



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74257, 78.04708



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86805, 78.08353



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87022, 78.08380



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87913, 78.08206



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92566, 78.06503



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93699, 78.05412



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95551, 78.04706



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95810, 78.04971



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95865, 78.04991



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95963, 78.04954



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96072, 78.04963



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96281, 78.05124



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96384, 78.05141



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96442, 78.05104



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96557, 78.04930



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96657, 78.04930



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12041, 78.13284



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.26533, 78.08428



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.26550, 78.08440



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.27173, 78.08098



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.29579, 78.07370



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29681, 78.07576



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.31143, 78.07861



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.31224, 78.07819



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.31368, 78.07448
